



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2009/2010

INSTITUTIONAL BASED PROGRAMME (APRIL SESSION)

EXAMINATION FOR THE DEGREE OF MASTER OF EDUCATION

ECT 529: DESIGN AND METHODS IN CHEMISTRY

DATE: THURSDAY 2ND SEPTEMBER 2010

TIME: 9.00 A.M. - 12.00 P.M.

INSTRUCTIONS: Answer any three questions.

1. “The process of development and instruction of science knowledge and chemistry in particular can be through induction and deduction.” As shown in the given diagram.

- (a) Outline the main steps used to ‘accept’ a chemical law.
- (b) Comment on the statement that “most instructional procedures in our schools follow the deductive approach.

- (c) With examples from the school chemistry syllabus, distinguish between a law and a theory.
- (d) What are the main differences in approach when teaching a chemical law and a theory at school level?

(23 marks)

2. A Resource plays an important role in the learning of chemistry.

- (a) Give the process that explains the role of a resource in chemistry instruction.
- (b) Using a chart as a resource: outline the qualities of a ‘good’ chart for chemistry instruction.
- (c) When teaching a topic “allotropes of carbon” teacher (A) uses charts and Teacher (B) uses models. Determine which teacher has made the better choice.
- (d) Apart from having a choice of a chart or a model to teach the topic on “allotropes of carbon”, suggest another possible resource.

(23 marks)

3. Micro-kits are becoming a common instructional resource in chemistry.

Compare the use of micro-kits and a conventional chemistry laboratory under the following headings.

- (a) The storage of chemicals
- (b) The economy in the use of chemicals
- (c) The diversity in experimental design
- (d) The rate of working on experiments

(23 marks)

4. “Molecular models form a valuable resource when teaching chemistry at school level.” Discuss,

- (a) The suitable type of model you would recommend to be used at:
 - (i) Form two level
 - (ii) Form four level

- (b) The main characteristics of a suitable model for chemistry instruction.
- (c) Some key parts (topics) of the chemistry syllabus can receive improved instruction using molecular models.
- (d) The main problems teachers face when acquiring molecular models.

(23 marks)

5. Explain the following using suitable examples.

- (a) There are two patterns of storing chemicals in a laboratory and both have similar advantages.
- (b) Using molecular models requires a careful choice to relate the type of model and the level of learners at school.
- (c) There are advantages of improvisation of a range of instructional resources by chemistry teachers.
- (d) Developing countries like Kenya have made some effort in producing some resources for chemistry teaching.

(23 marks)
